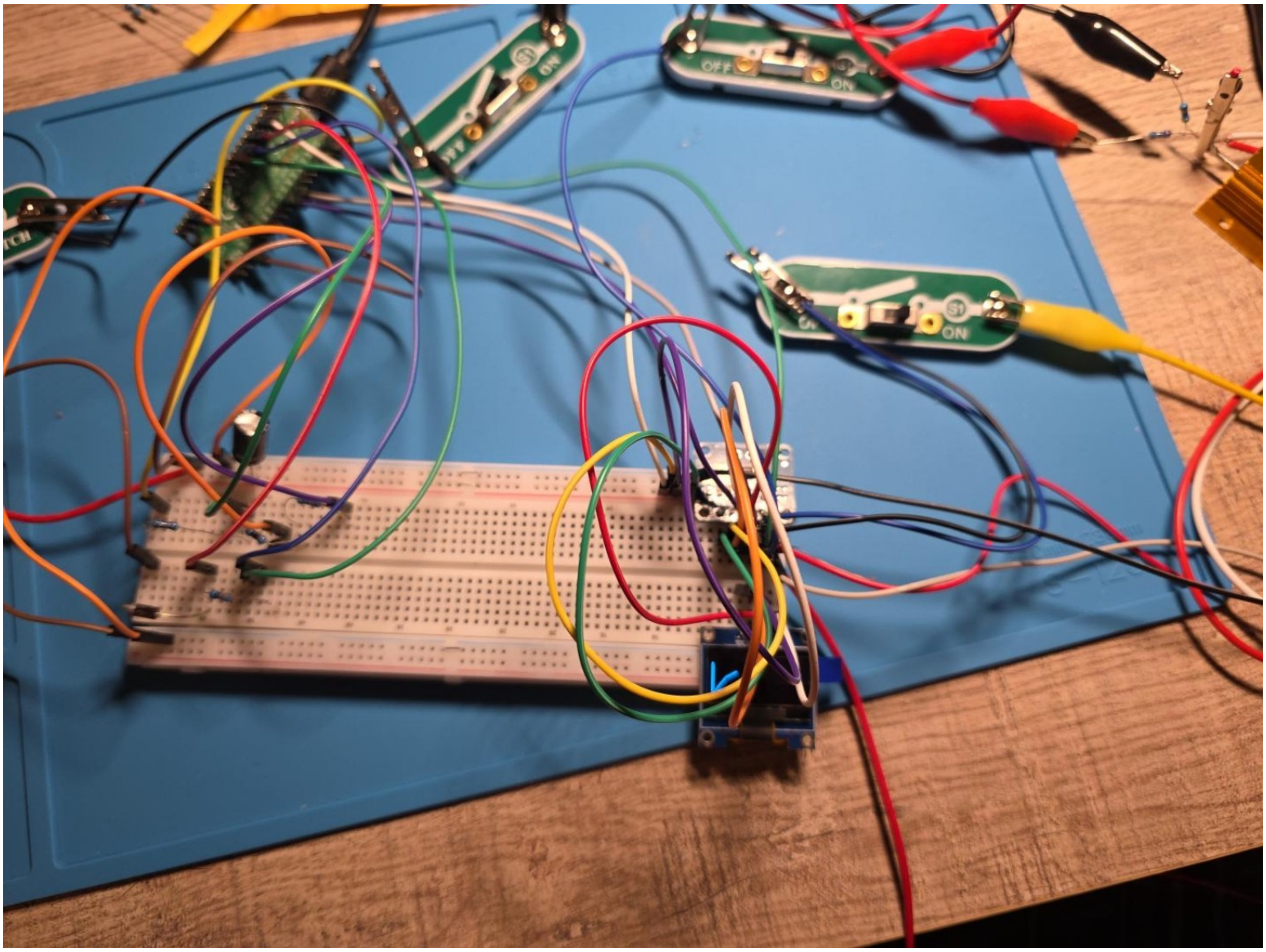
EE Lab Tool - Complete Sanitized Photo Appendix

All 26 photographs supplied for the final project update. Screen images were cropped to remove browser chrome, tabs/bookmarks, local IP information, laptop branding, and unrelated interface content.

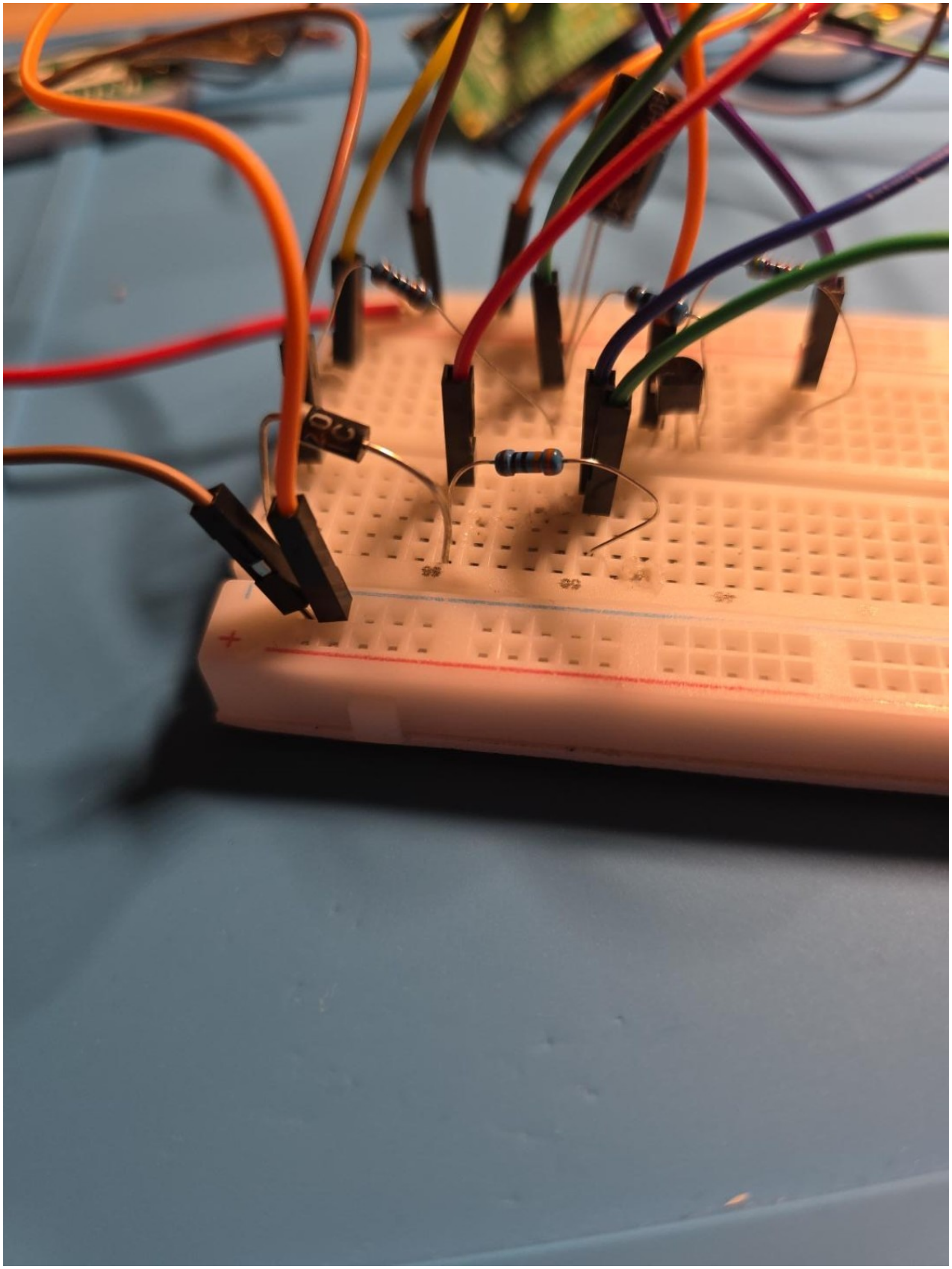
Figure 1



Overall prototype overview showing the complete Pico 2 W EE Lab Tool on the work mat, including the breadboard, external switch modules, OLED, and surrounding test wiring.

Privacy treatment: none required. Source file: image-1787893448668.jpg.

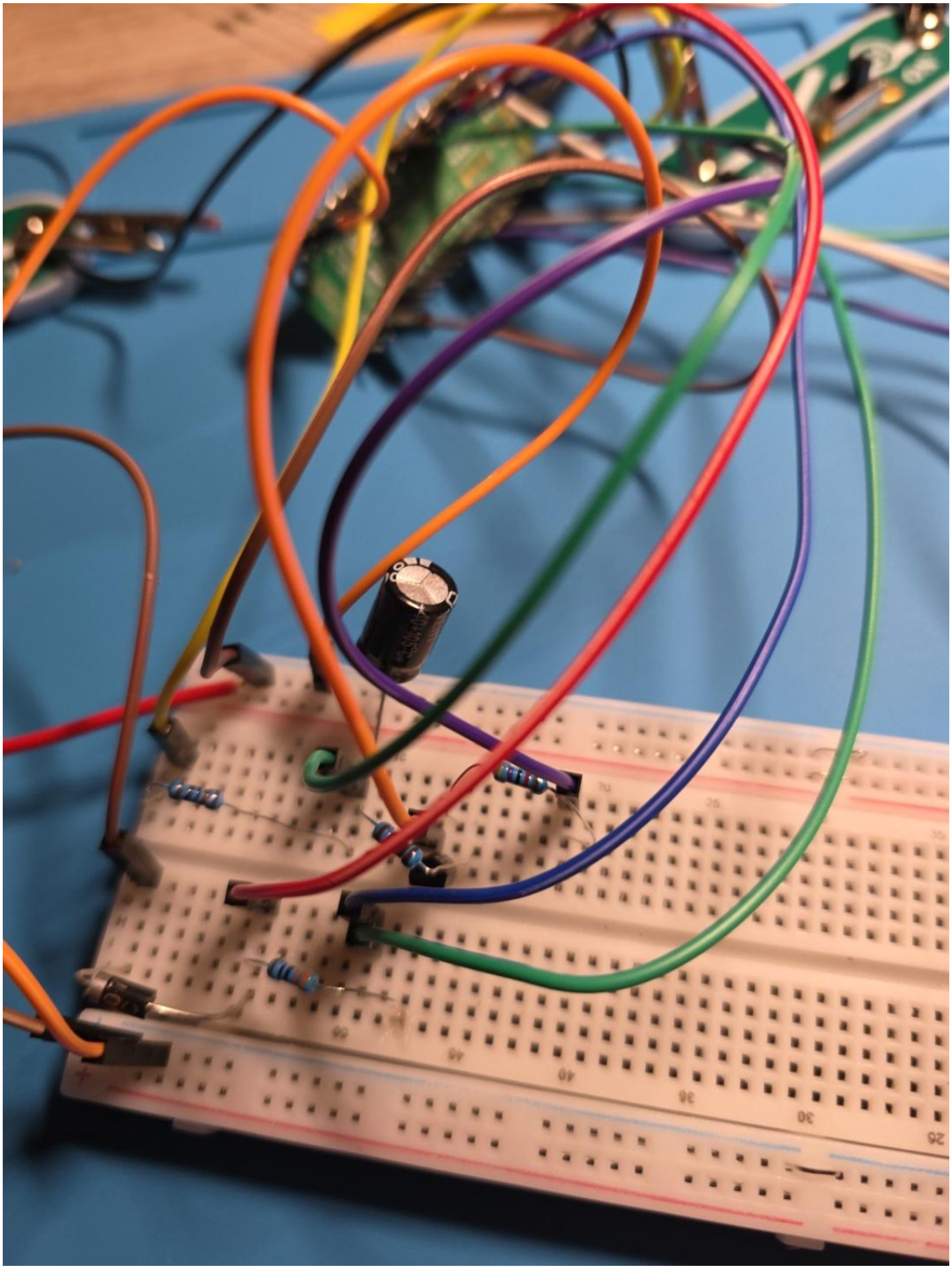
Figure 2



Close-up of the low-current breadboard test area, preserving component placement and jumper routing around the transistor/resistor network.

Privacy treatment: none required. Source file: image-1787893452363.jpg.

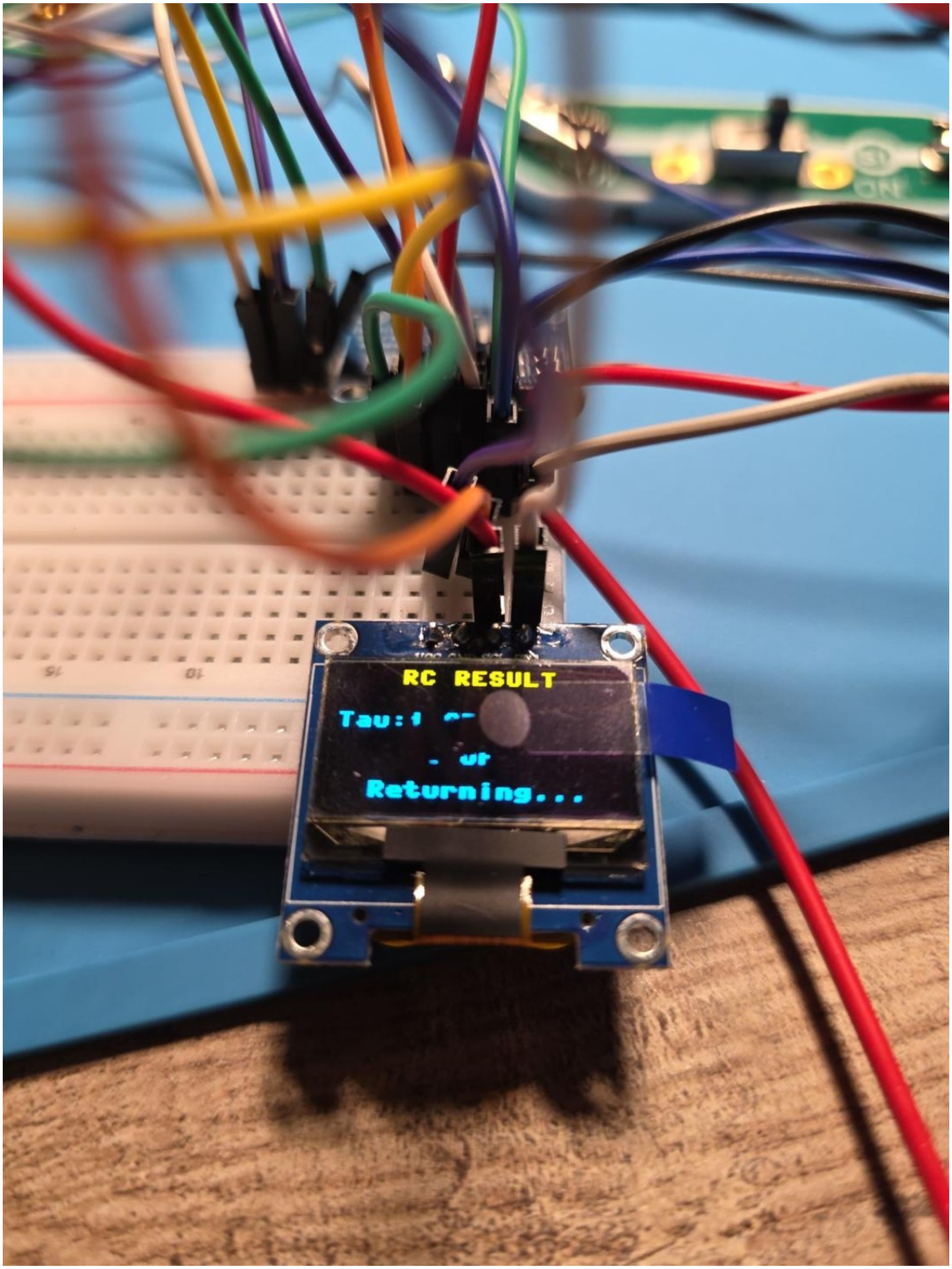
Figure 3



RC analyzer section close-up showing the 1000 uF electrolytic capacitor, resistors, transistor, and ADC/control wiring.

Privacy treatment: none required. Source file: image-1787893461592.jpg.

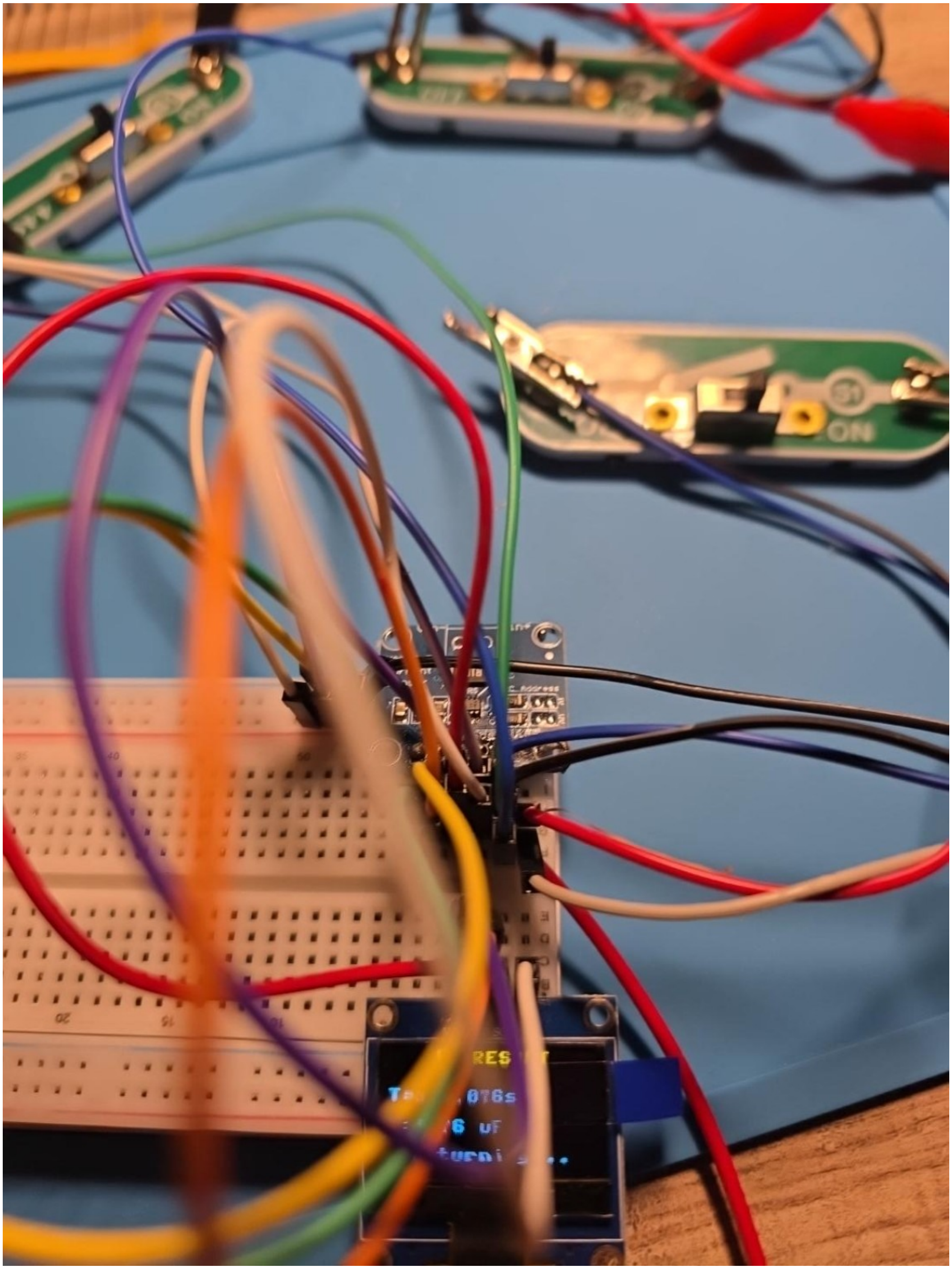
Figure 4



OLED RC Analyzer result screen documenting a successful measured time constant and capacitance result.

Privacy treatment: none required. Source file: image-1787893464888.jpg.

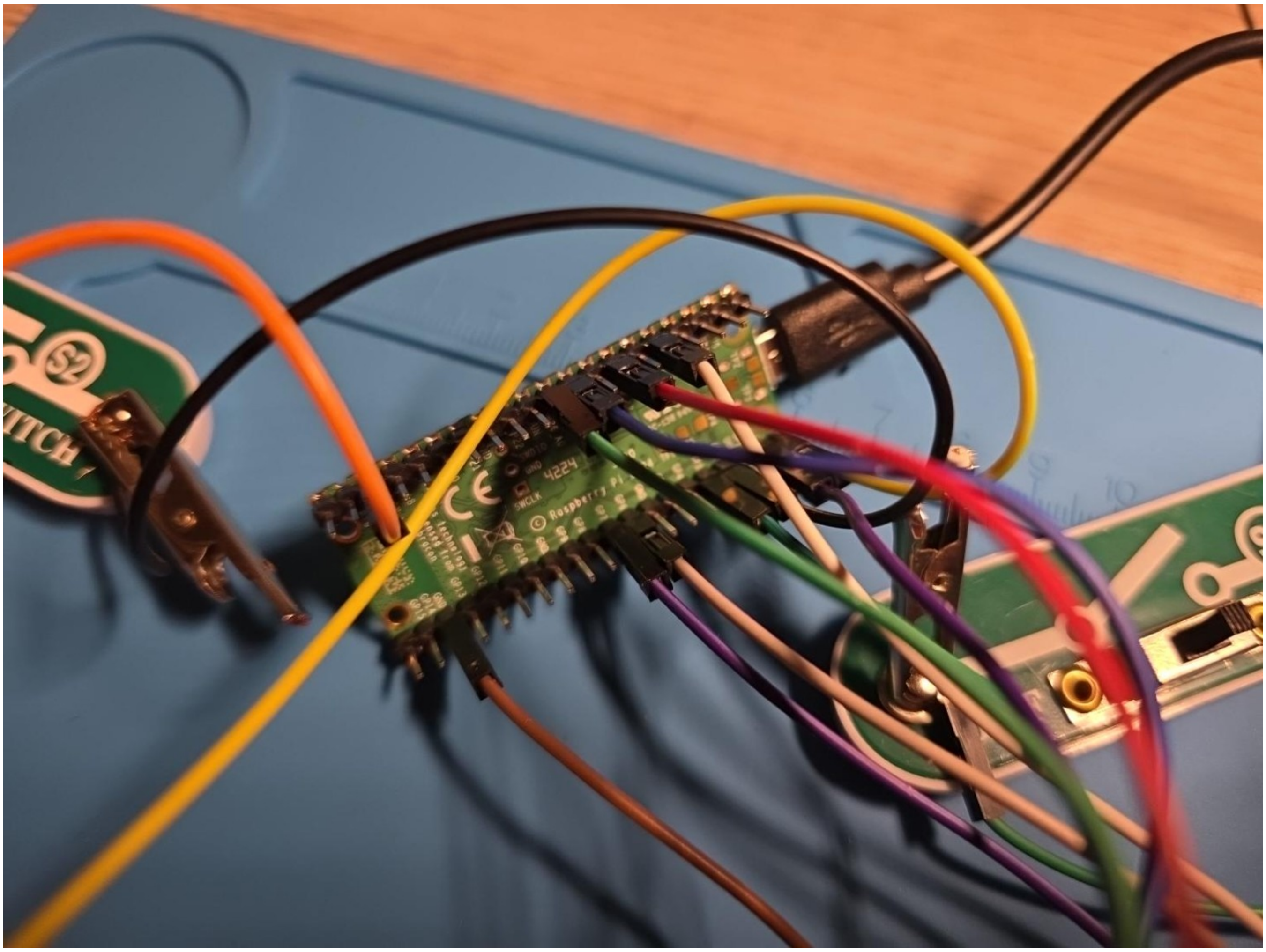
Figure 5



Top-side close-up of the Pico 2 W, INA219/OLED interconnections, and dense shared-bus/control wiring.

Privacy treatment: none required. Source file: image-1787893467914.jpg.

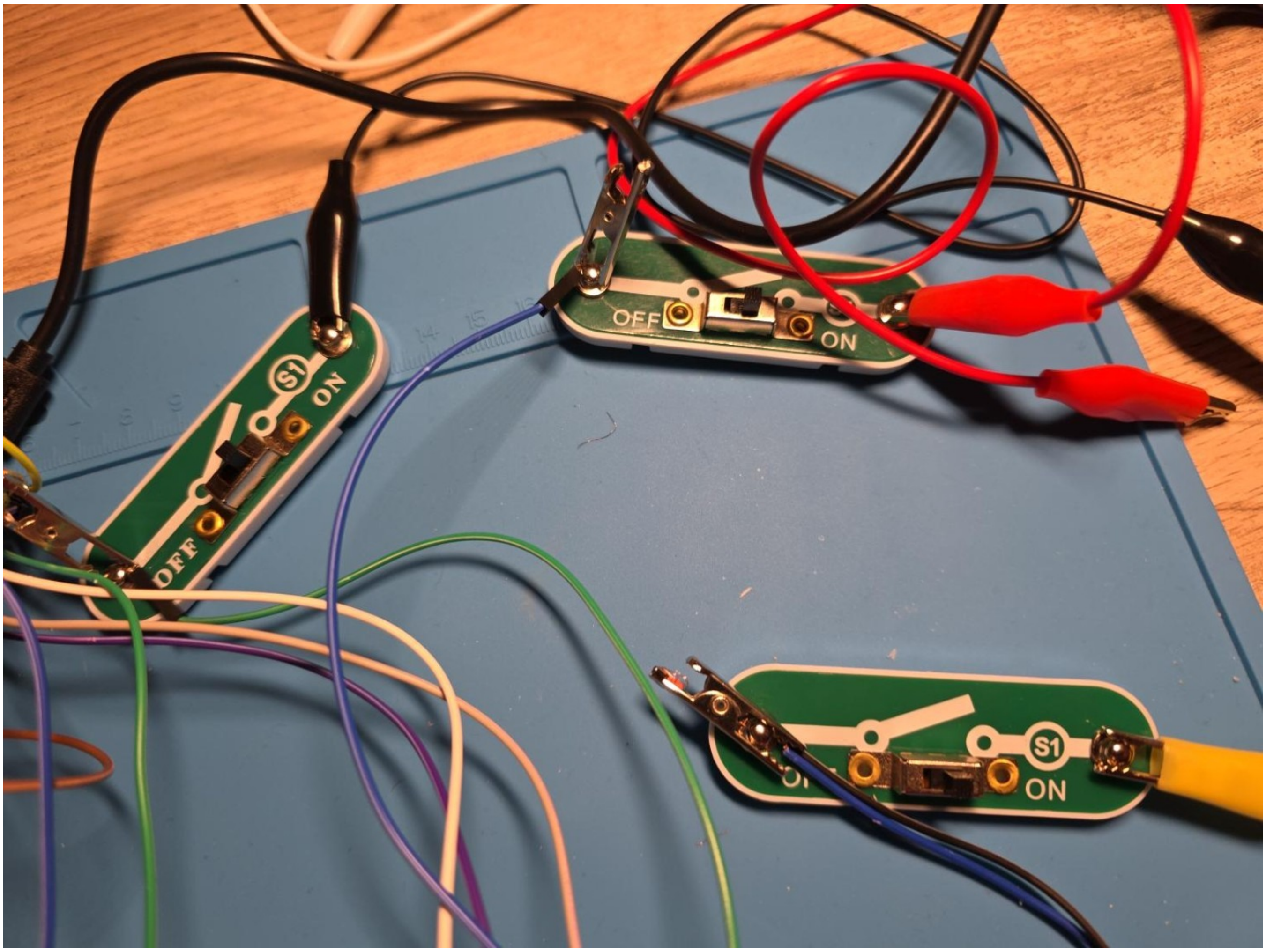
Figure 6



Raspberry Pi Pico 2 W controller close-up documenting the header connections used by the integrated instrument.

Privacy treatment: none required. Source file: image-1787893481899.jpg.

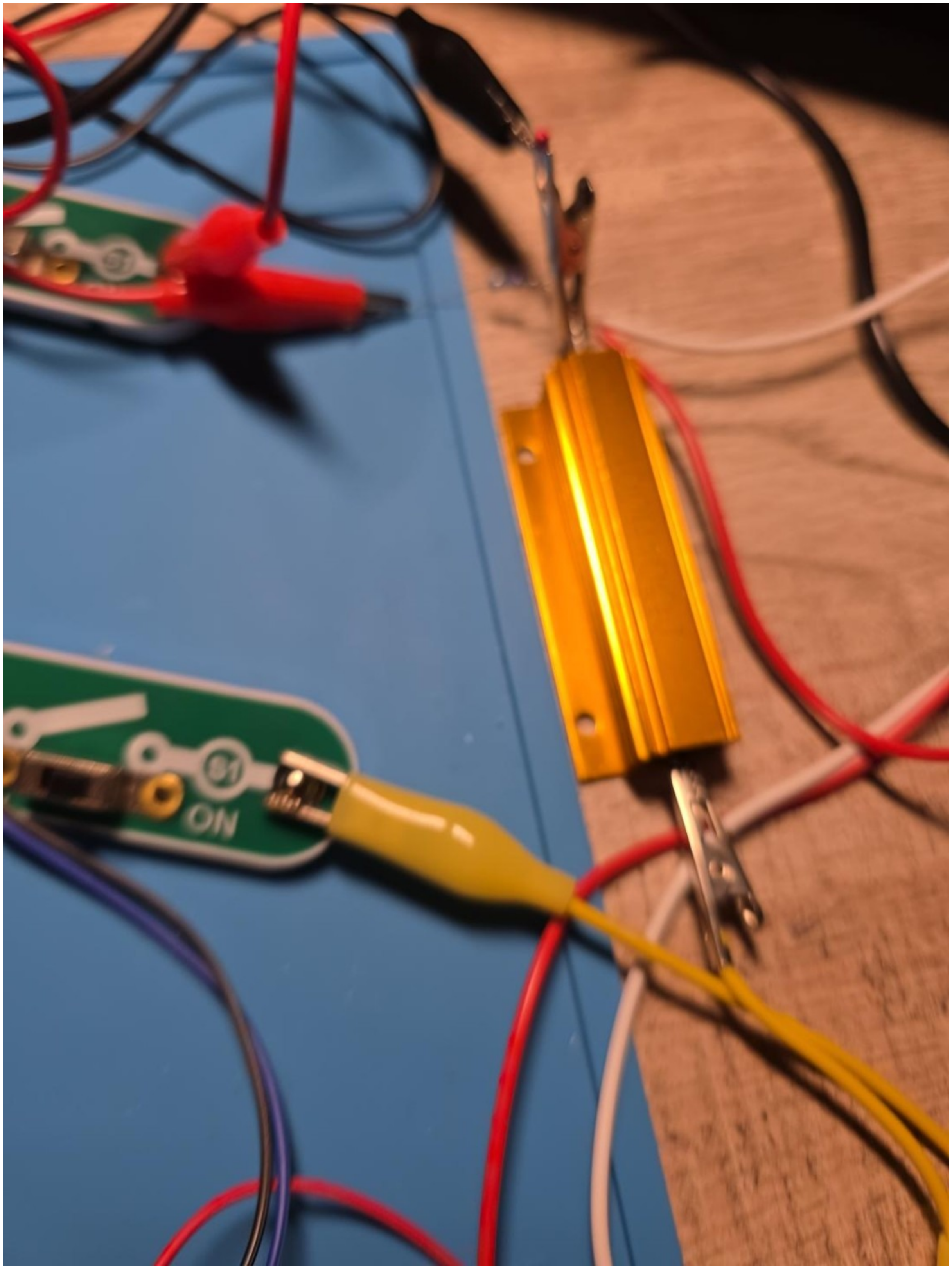
Figure 7



Manual ON/OFF switch modules and alligator-clip wiring used as physical circuit selectors/bypass elements during testing.

Privacy treatment: none required. Source file: image-1787893486860.jpg.

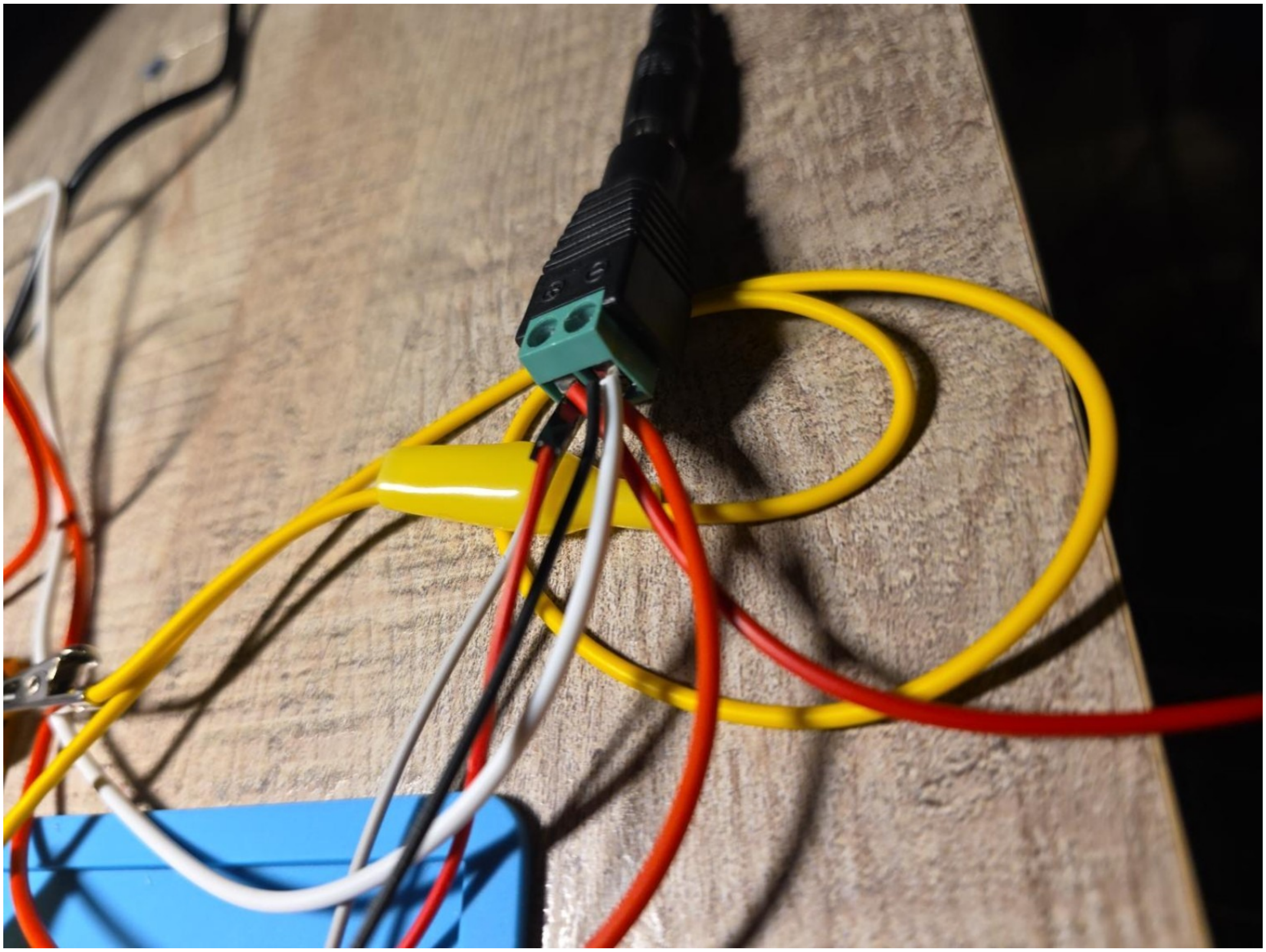
Figure 8



10 ohm, 100 W aluminum power resistor used as the principal high-current load.

Privacy treatment: none required. Source file: image-1787893496016.jpg.

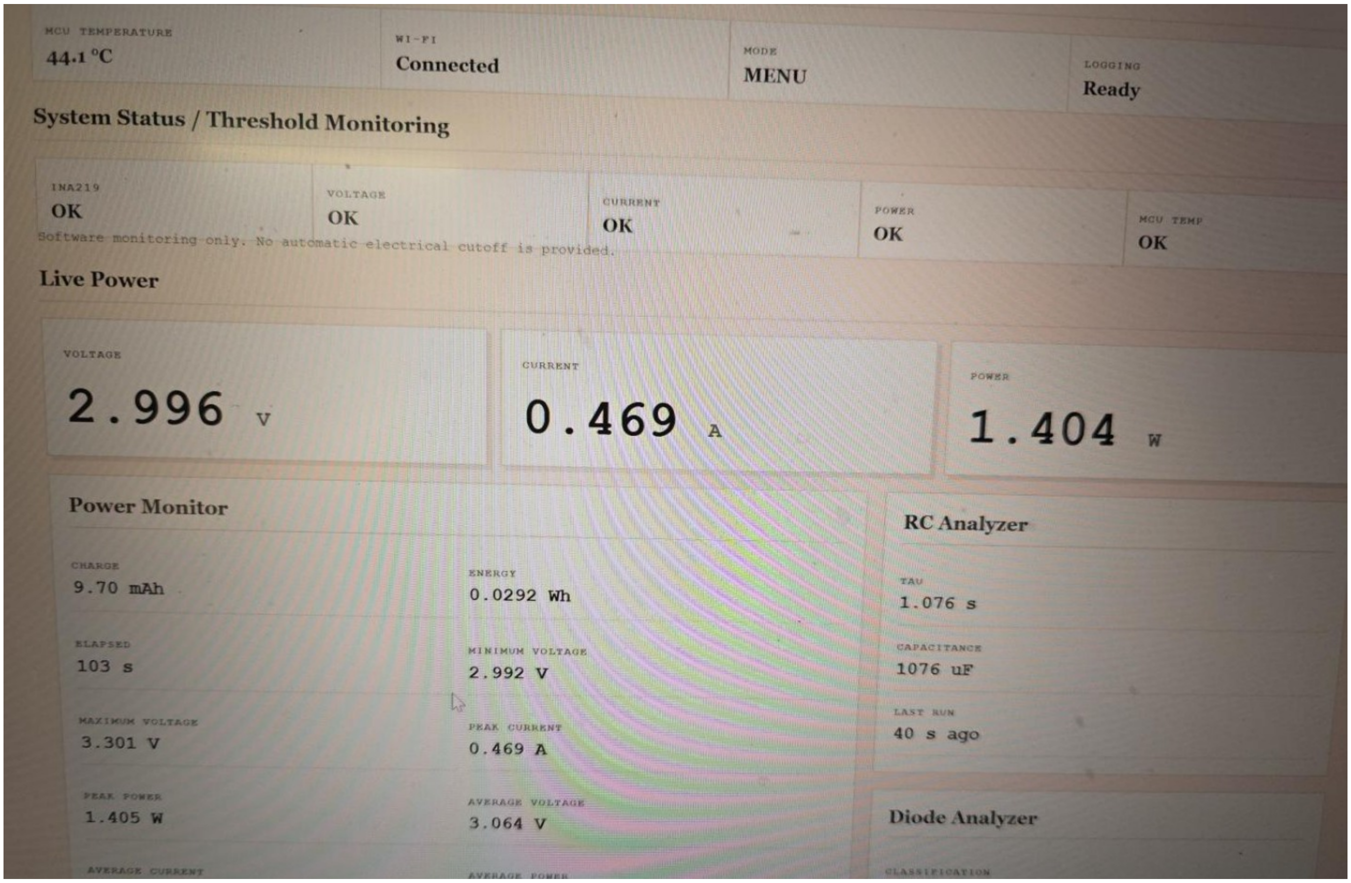
Figure 9



Screw-terminal power-harness breakout and multi-conductor wiring used for detachable source/load connections.

Privacy treatment: none required. Source file: image-1787893499406.jpg.

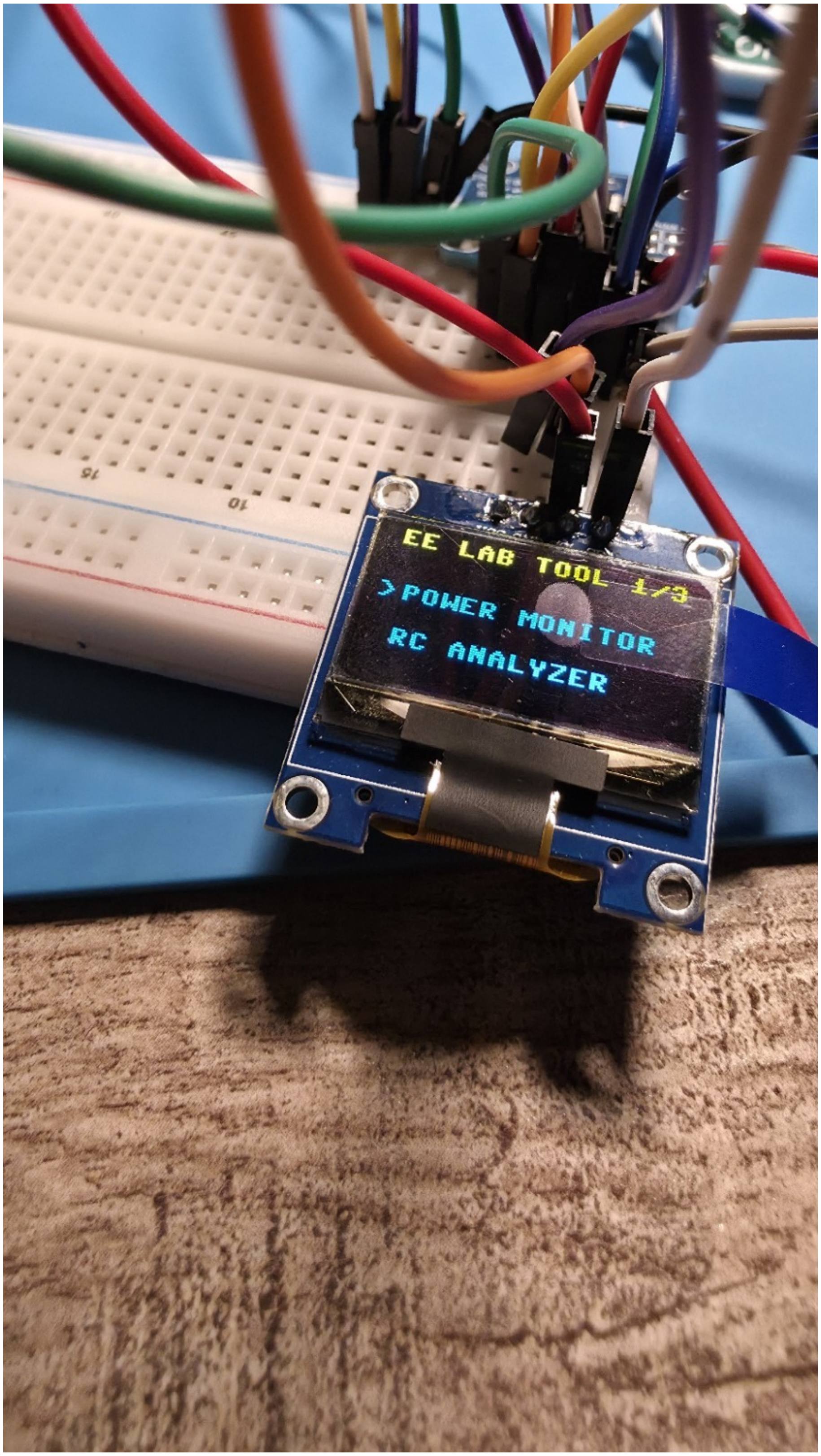
Figure 10



Sanitized dashboard overview showing MCU temperature, Wi-Fi status, threshold monitoring, live power, Power Monitor statistics, and RC Analyzer results.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893506140.jpg.

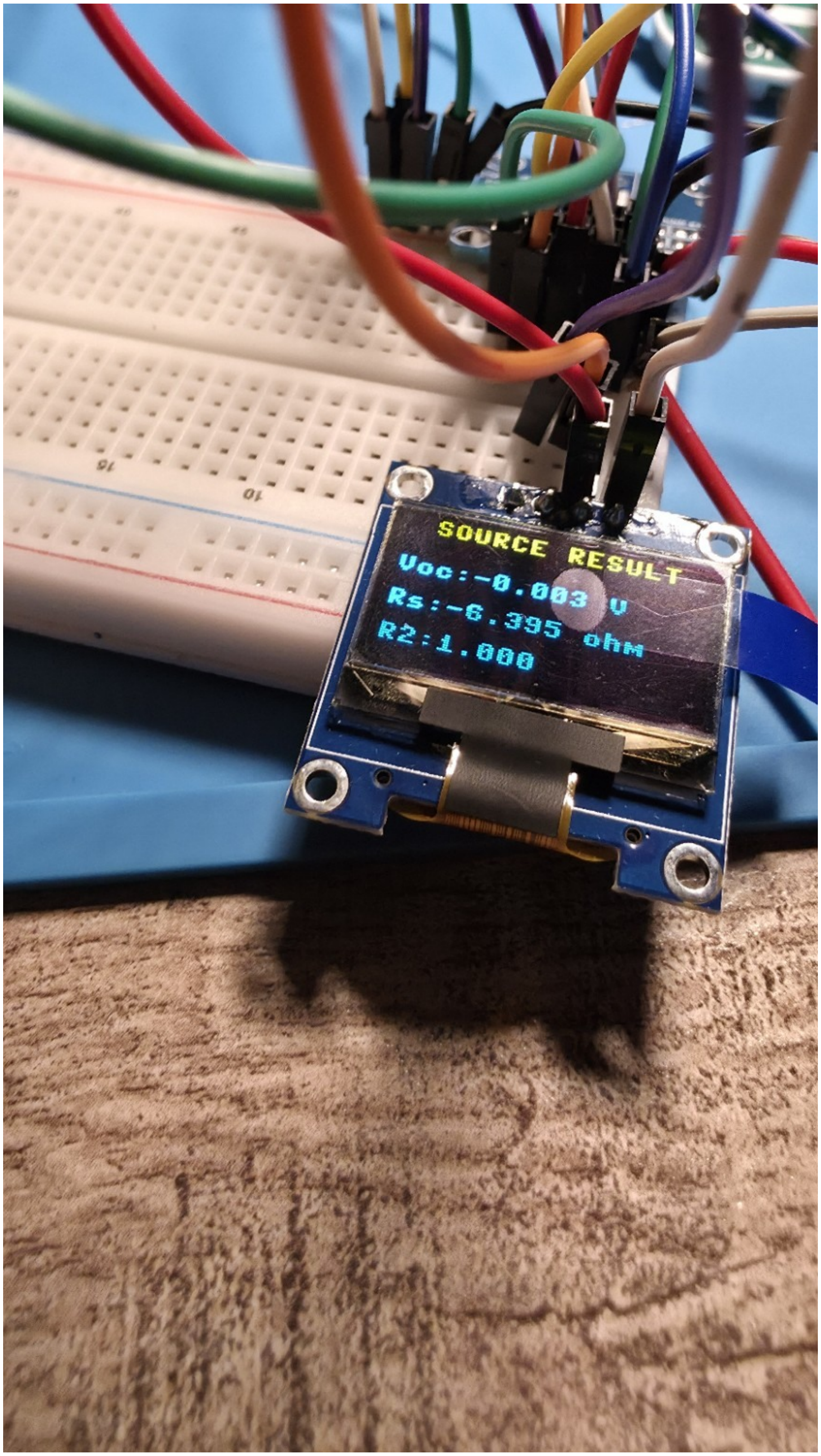
Figure 11



OLED main-menu page showing the unified EE LAB TOOL interface and analyzer selection.

Privacy treatment: none required. Source file: 1000017942.jpg.

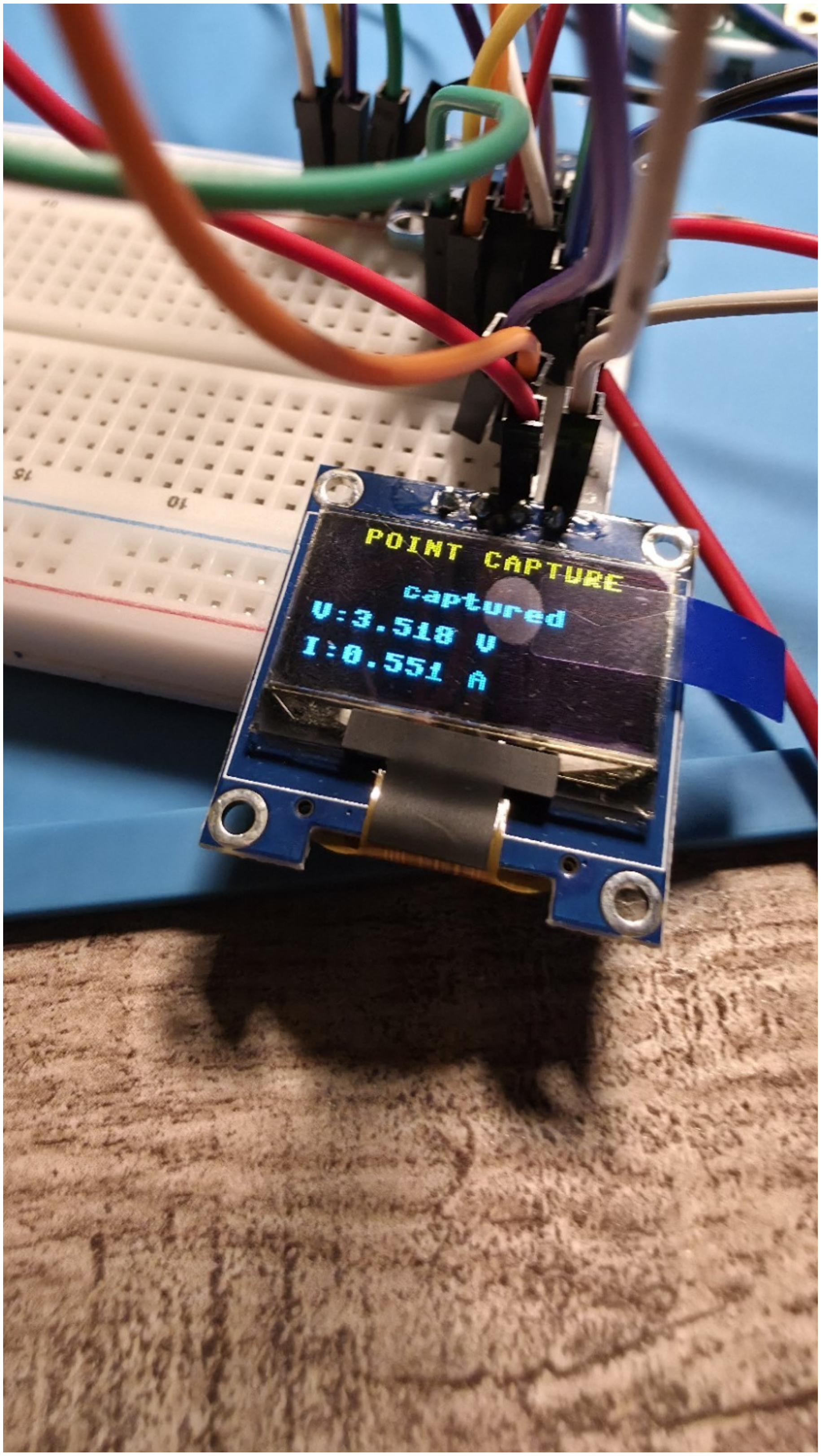
Figure 12



OLED Source Test result screen. The photographed negative source-resistance result is retained as a validation issue, not accepted as a physical source model.

Privacy treatment: none required. Source file: 1000017943.jpg.

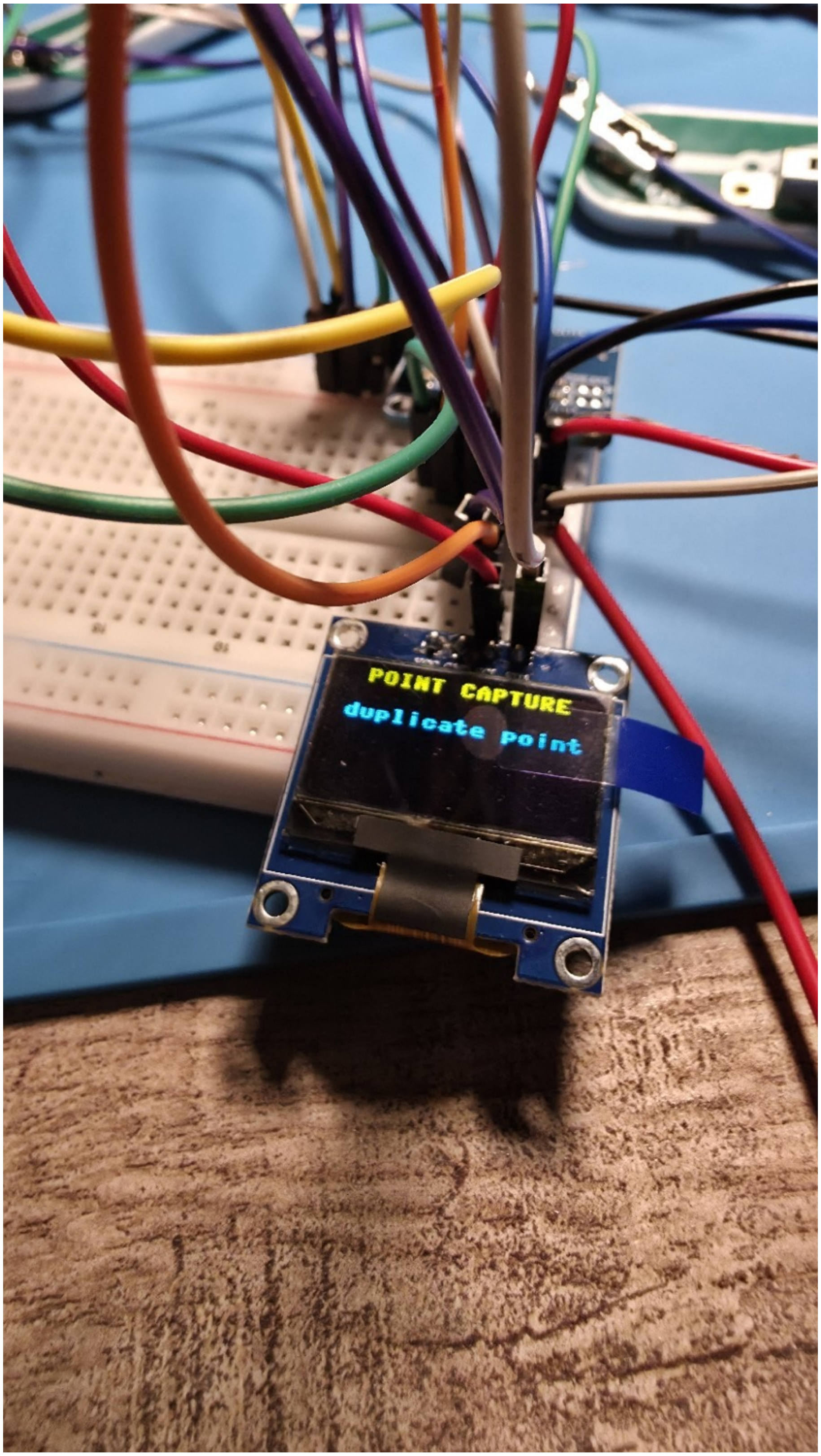
Figure 13



OLED Point Capture confirmation with the measured voltage and current for an accepted source-test point.

Privacy treatment: none required. Source file: 1000017944.jpg.

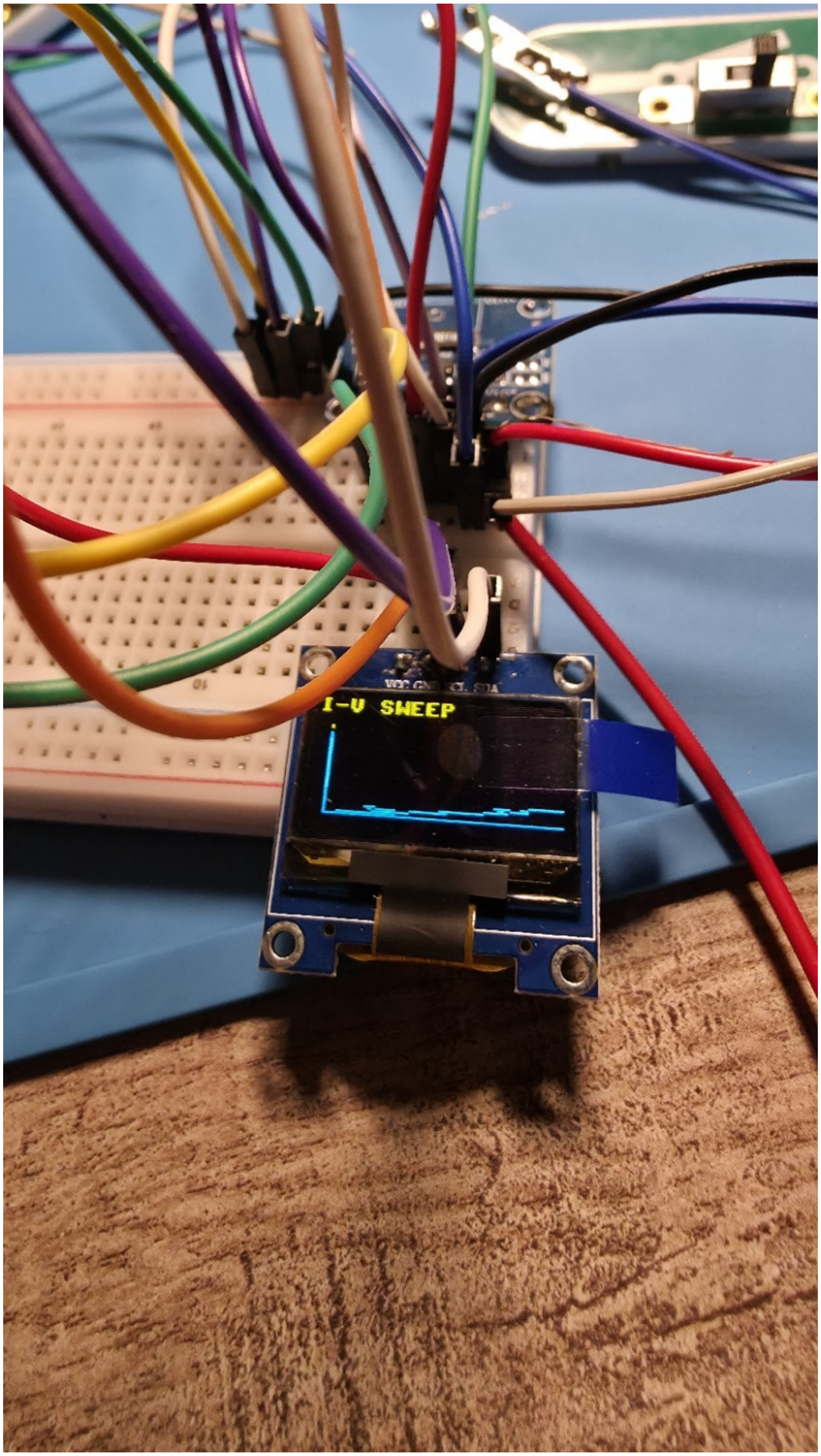
Figure 14



OLED duplicate-point rejection message demonstrating source-test data-quality screening.

Privacy treatment: none required. Source file: 1000017945.jpg.

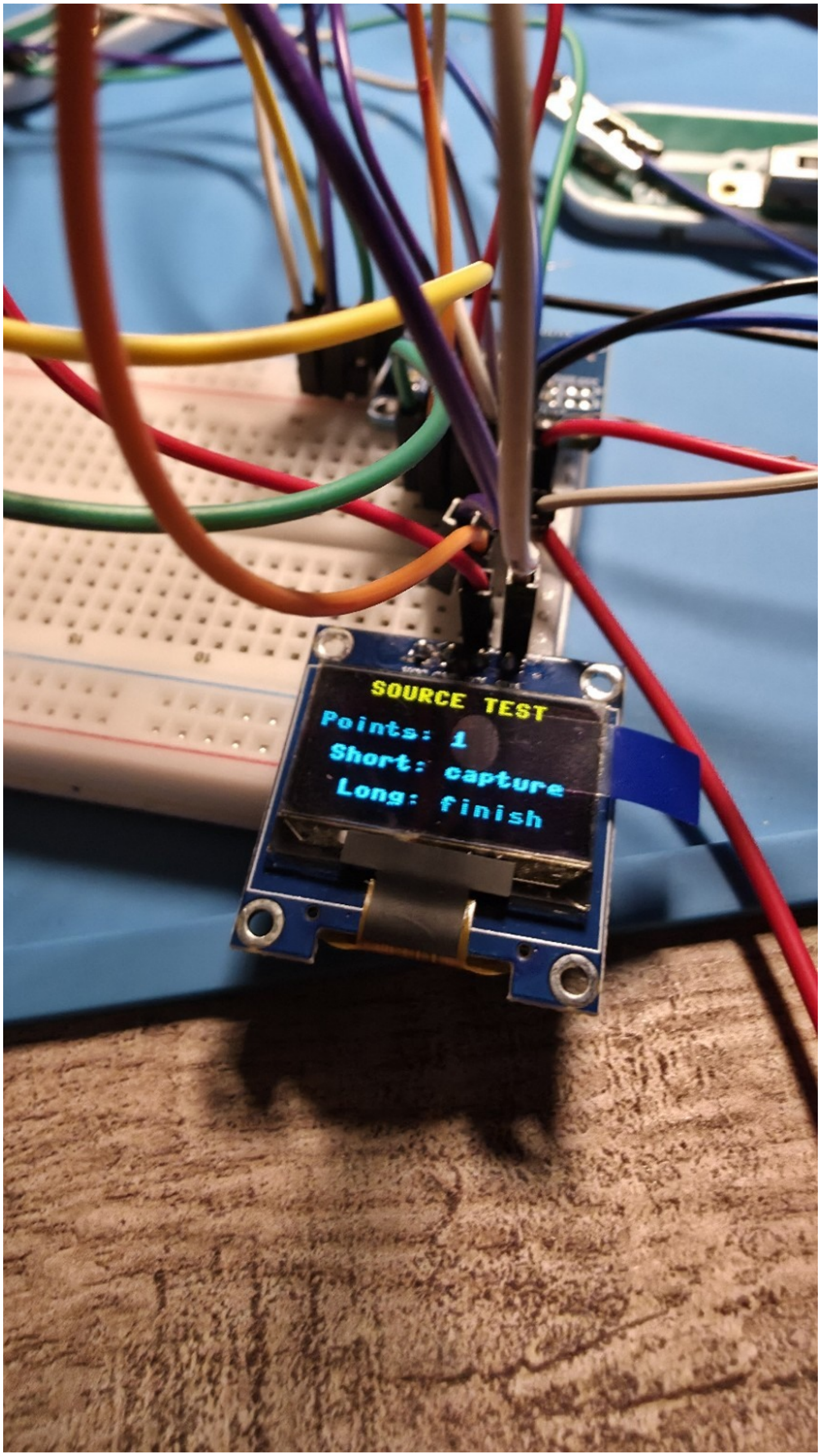
Figure 15



OLED I-V sweep visualization from the Diode Analyzer.

Privacy treatment: none required. Source file: 1000017947.jpg.

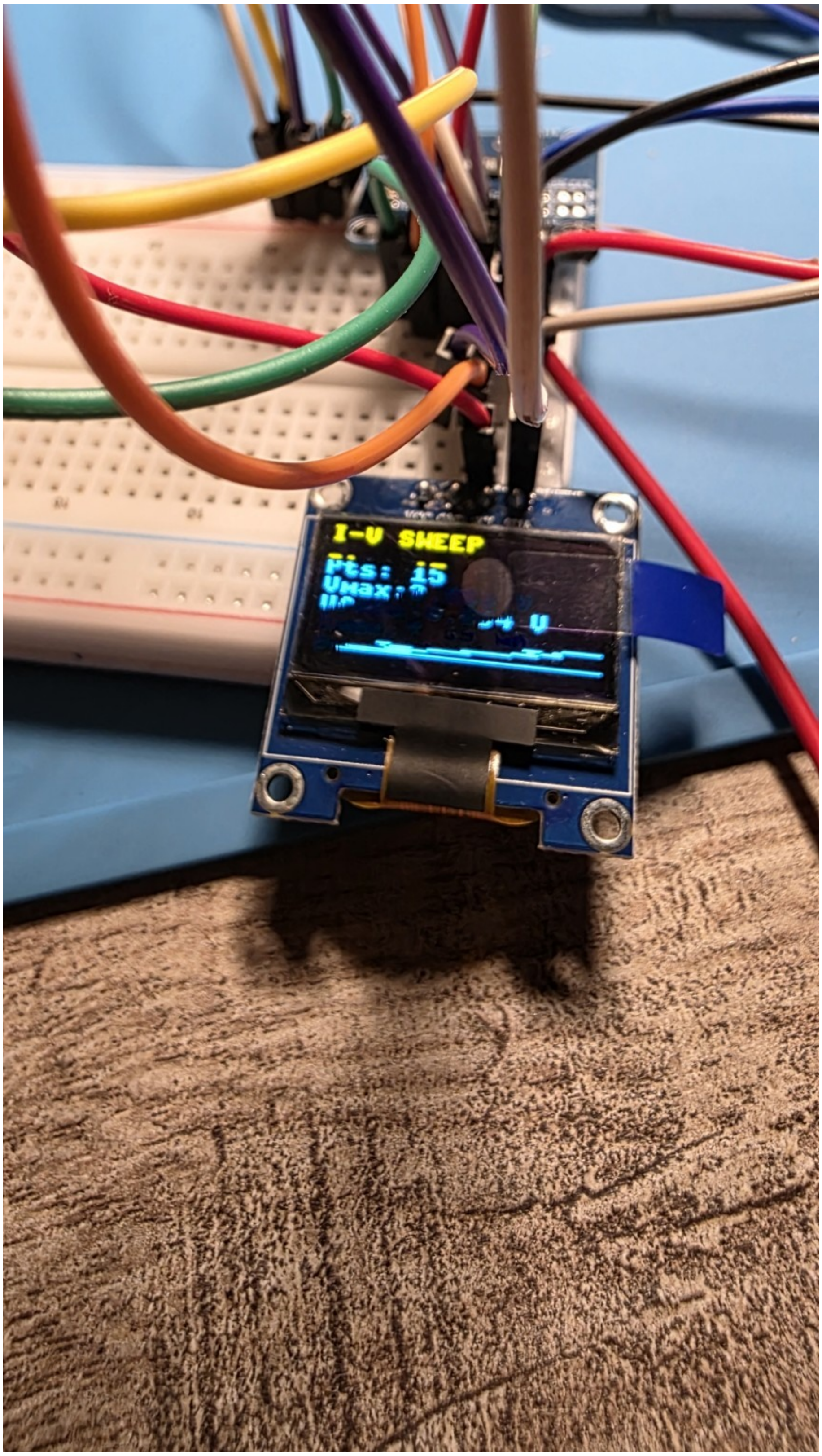
Figure 16



OLED Source Test collection screen showing the current point count and short/long press controls.

Privacy treatment: none required. Source file: 1000017946.jpg.

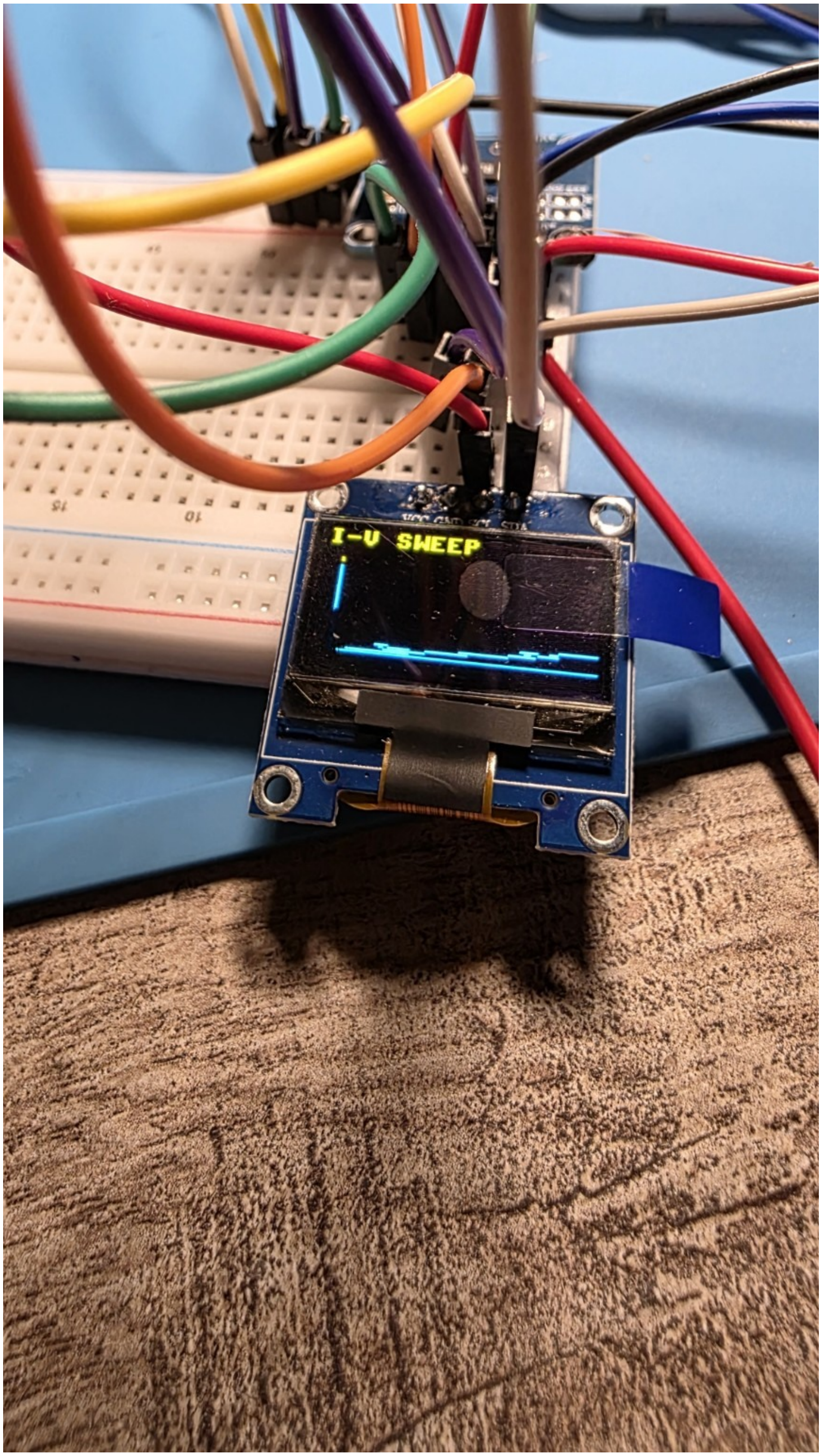
Figure 17



OLED I-V sweep numerical/statistical page documenting the captured sweep summary.

Privacy treatment: none required. Source file: 1000017940.jpg.

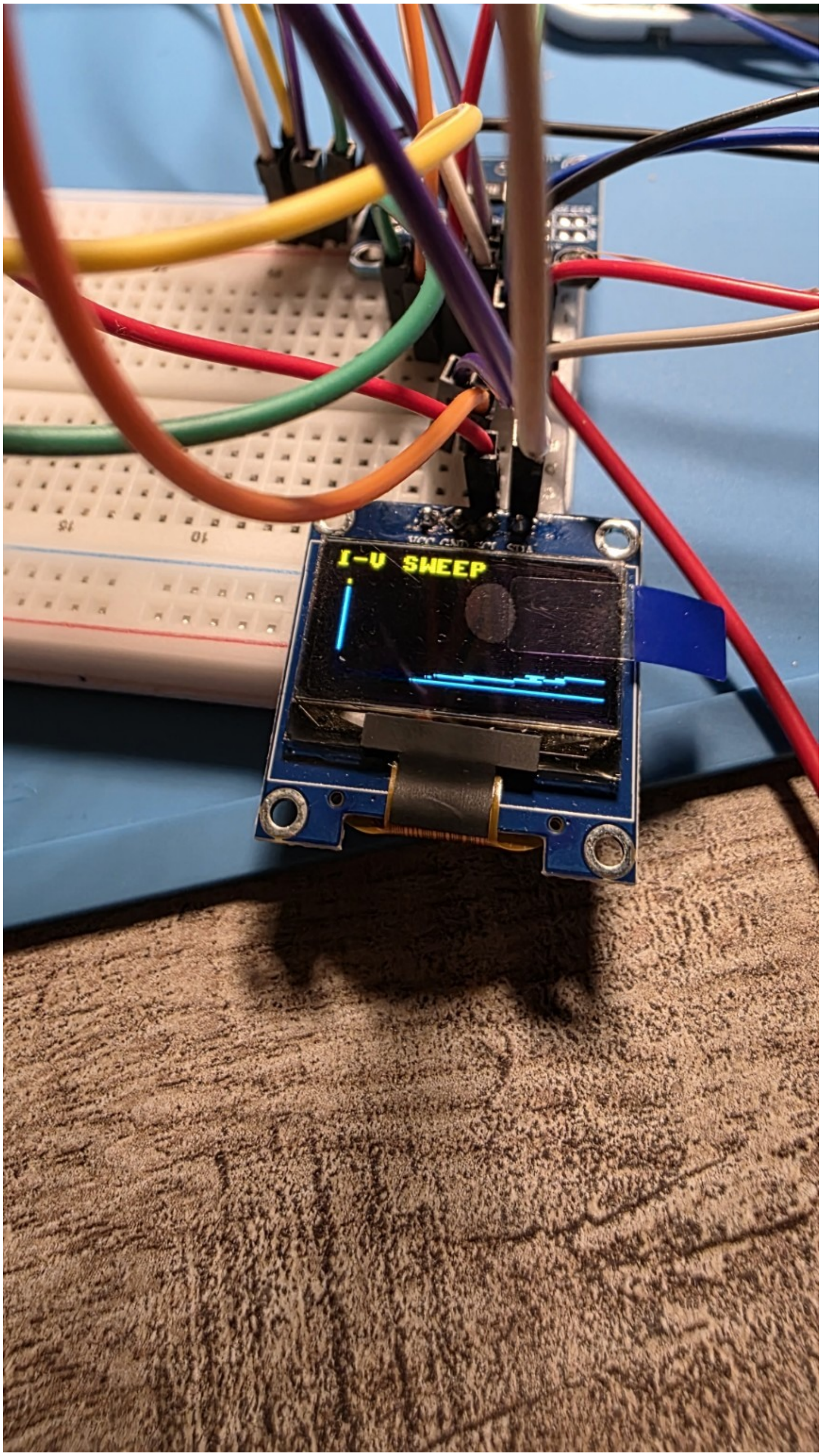
Figure 18



OLED I-V sweep graph, alternate captured frame.

Privacy treatment: none required. Source file: 1000017941.jpg.

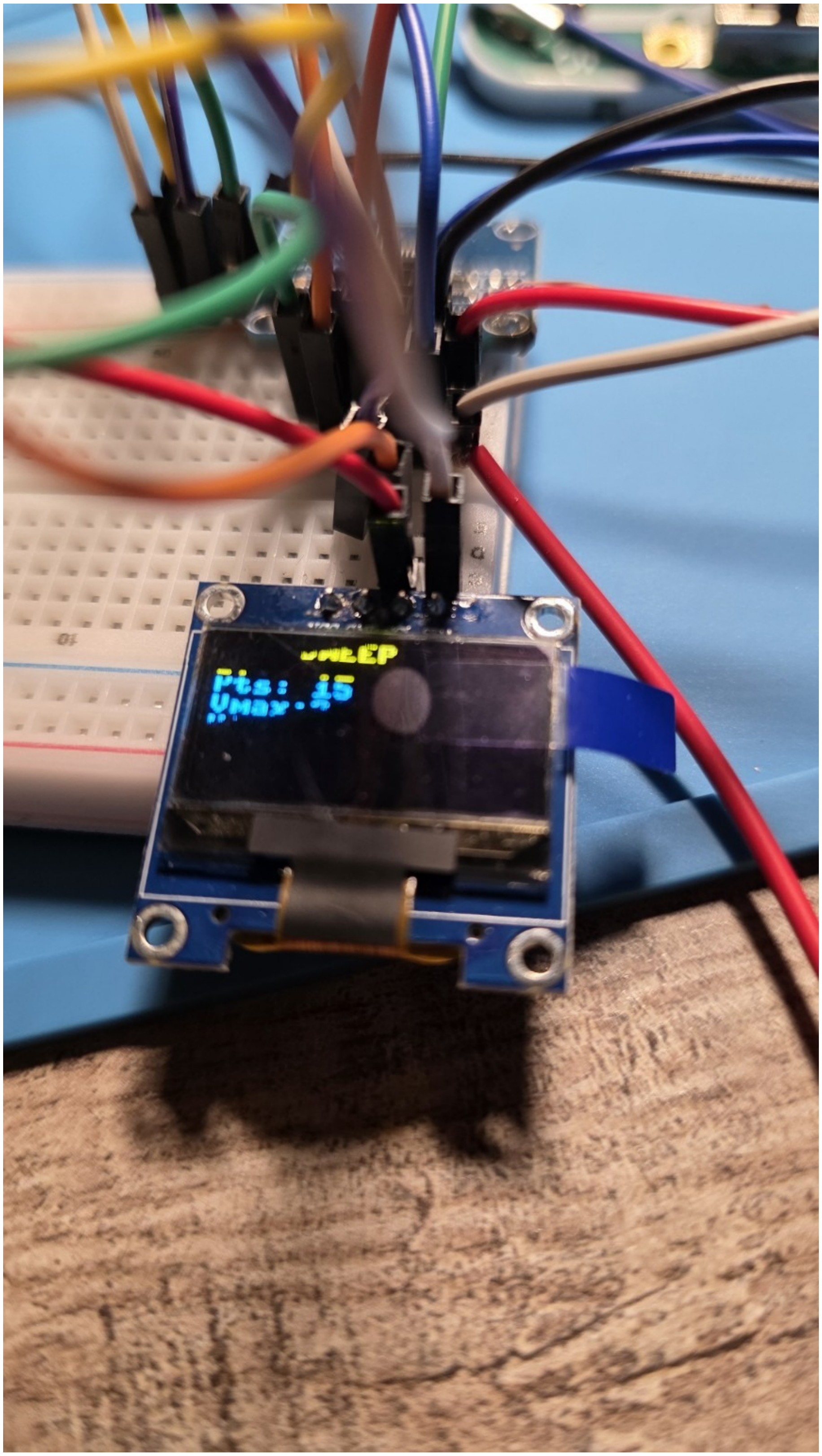
Figure 19



OLED I-V sweep graph, alternate captured frame preserving the displayed curve.

Privacy treatment: none required. Source file: 1000017939.jpg.

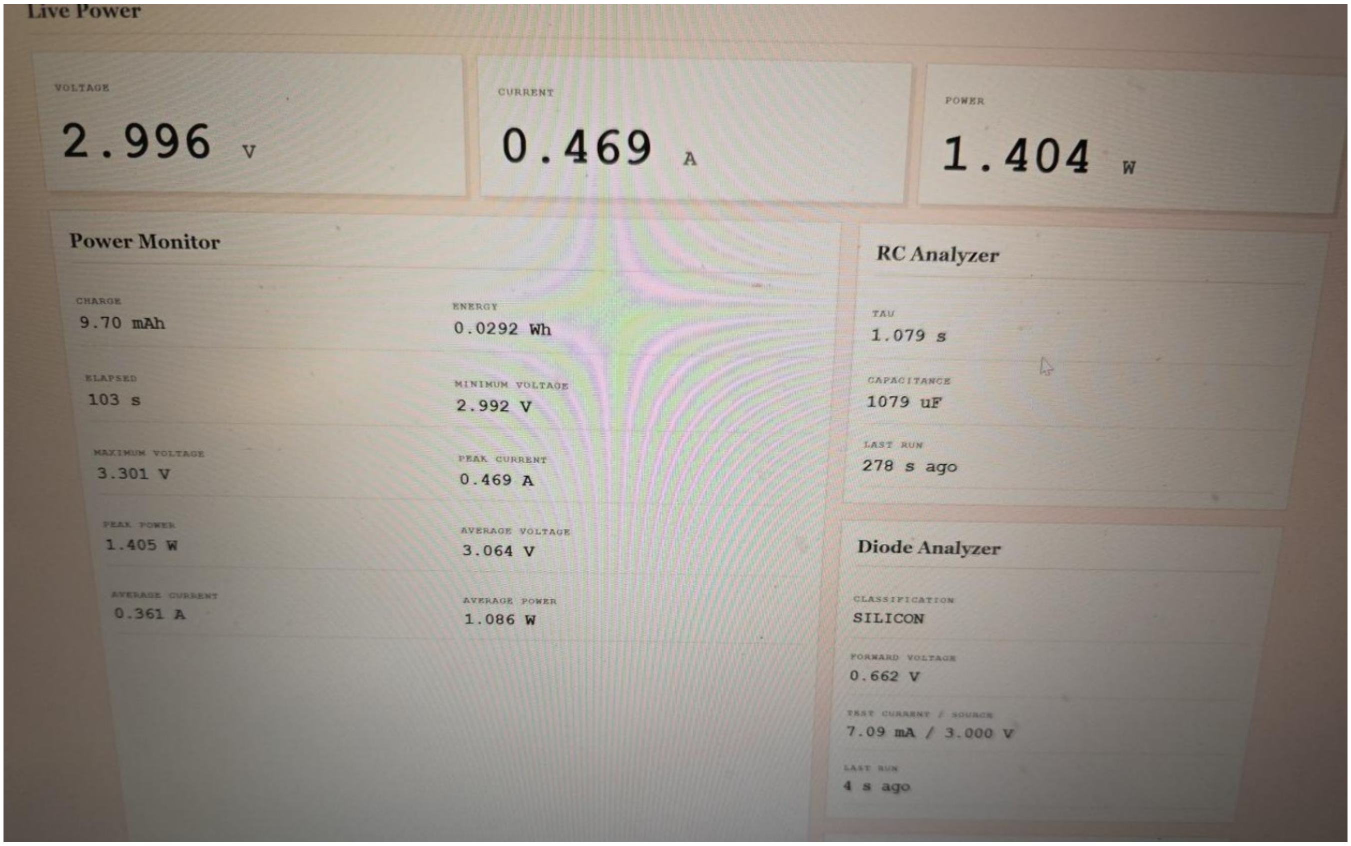
Figure 20



OLED I-V sweep statistics/graph frame documenting the embedded visualization workflow.

Privacy treatment: none required. Source file: 1000017938.jpg.

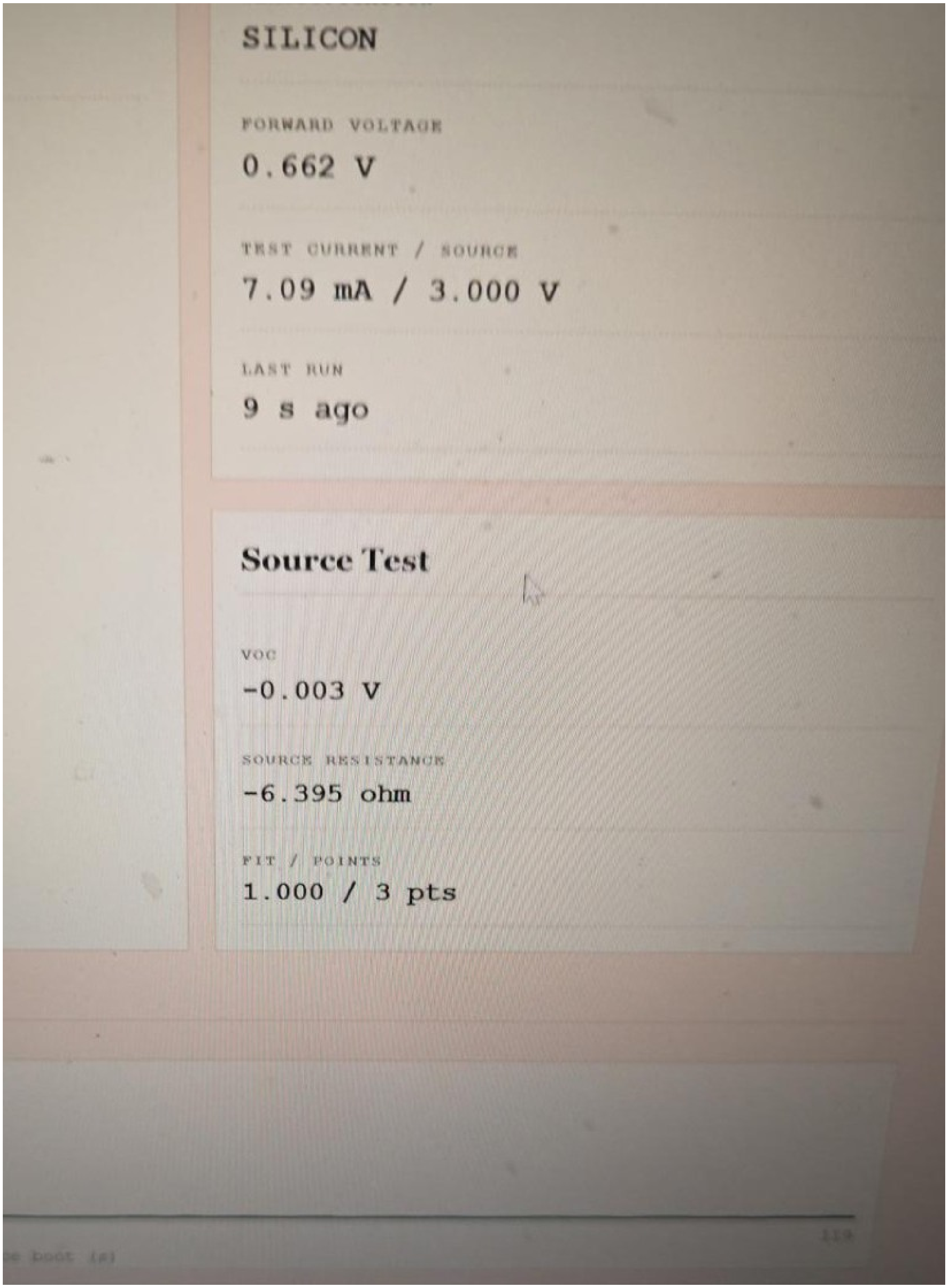
Figure 21



Sanitized dashboard view showing live voltage/current/power, Power Monitor session statistics, RC Analyzer results, and the silicon-diode result.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893815724.jpg.

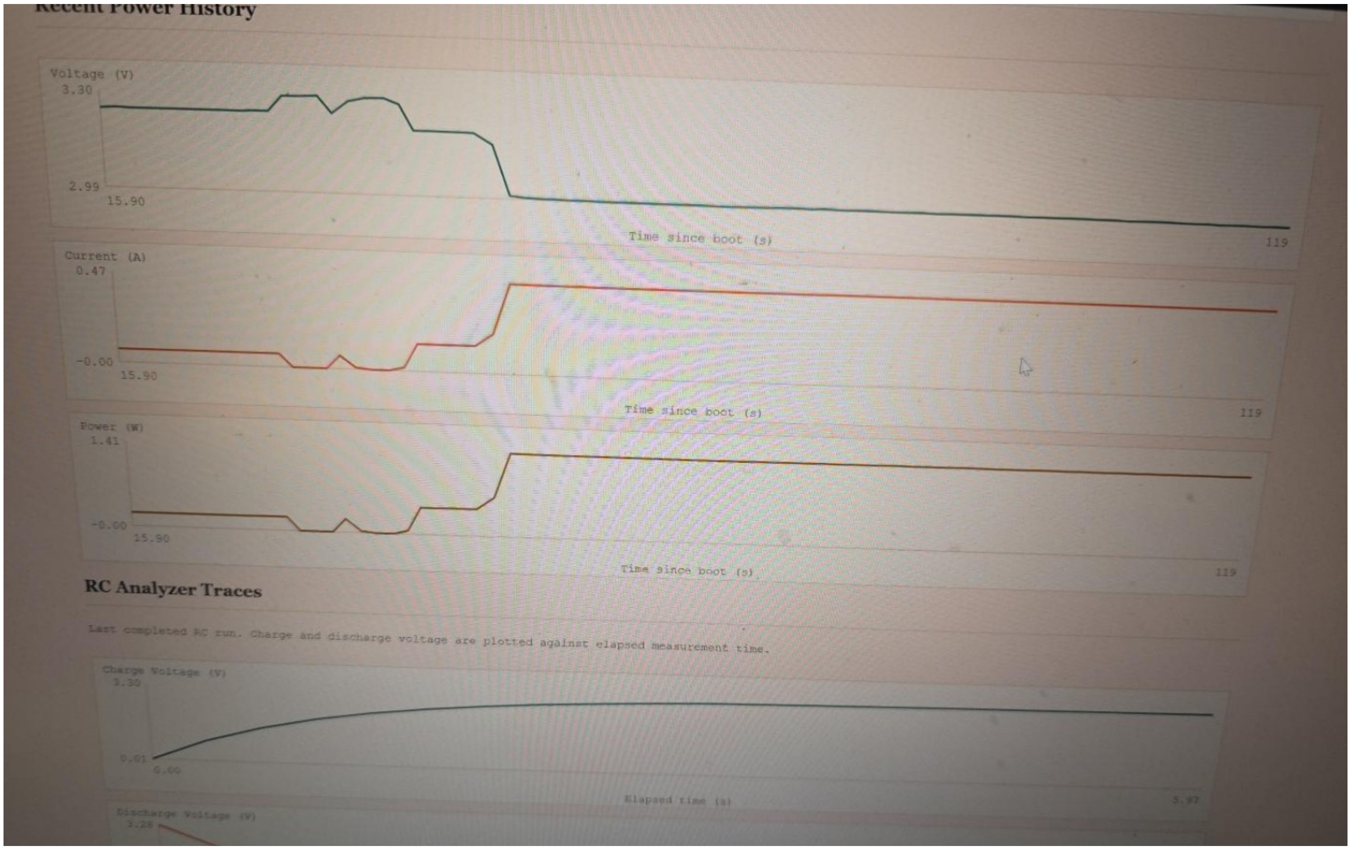
Figure 22



Sanitized dashboard close-up showing the silicon-diode result and the three-point Source Test fit that produced a negative resistance and therefore requires validation.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893821029.jpg.

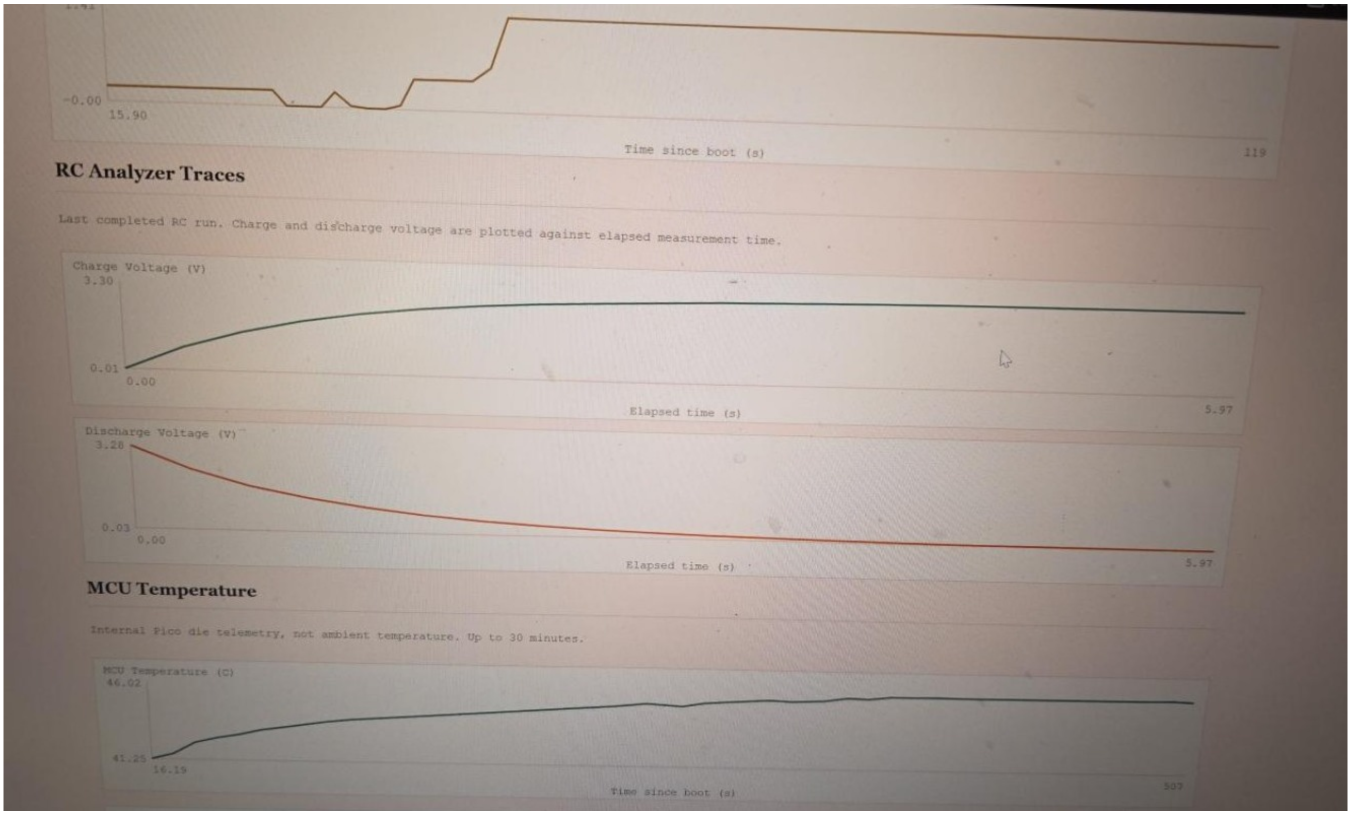
Figure 23



Sanitized dashboard charts showing recent voltage/current/power history and RC charge/discharge traces.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893828096.jpg.

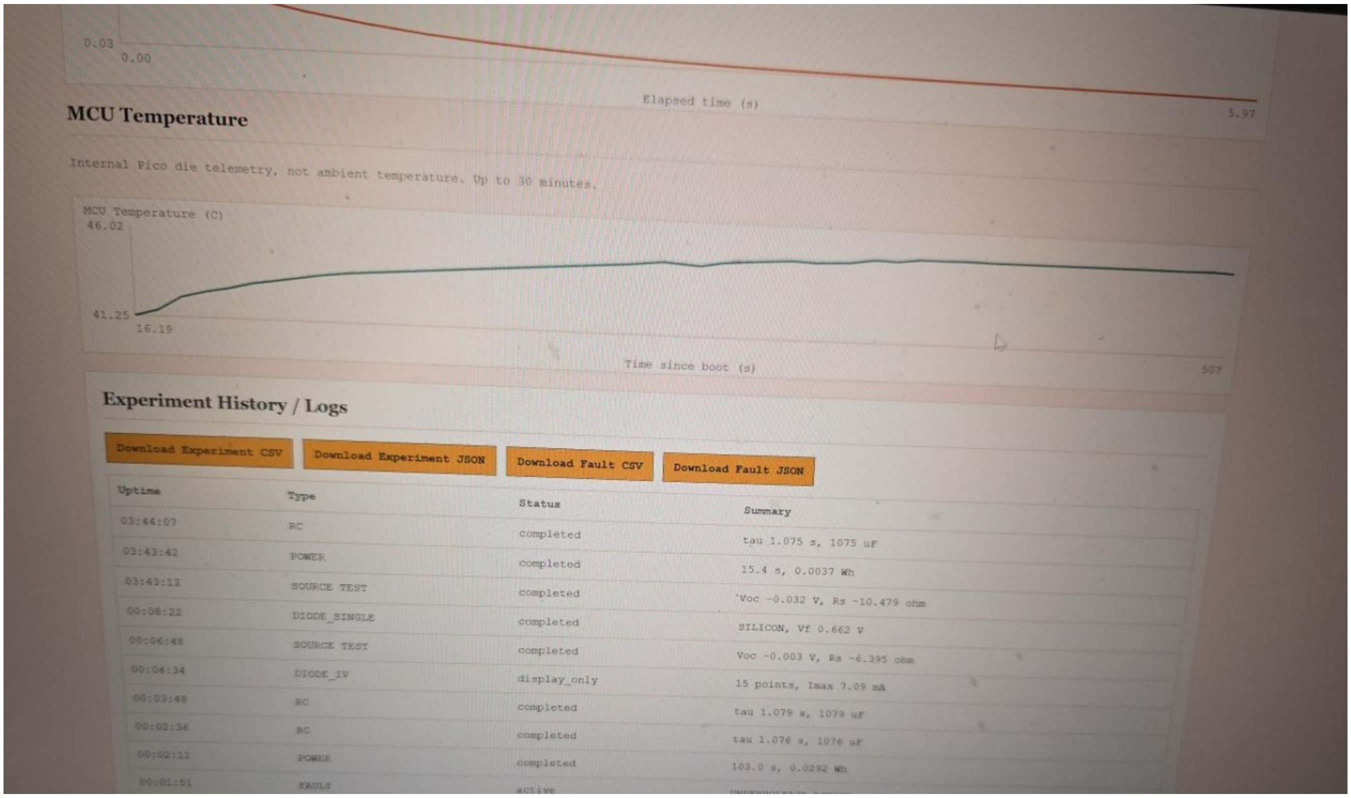
Figure 24



Sanitized dashboard view showing RC traces, MCU internal-die temperature history, and experiment/fault export controls.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893835341.jpg.

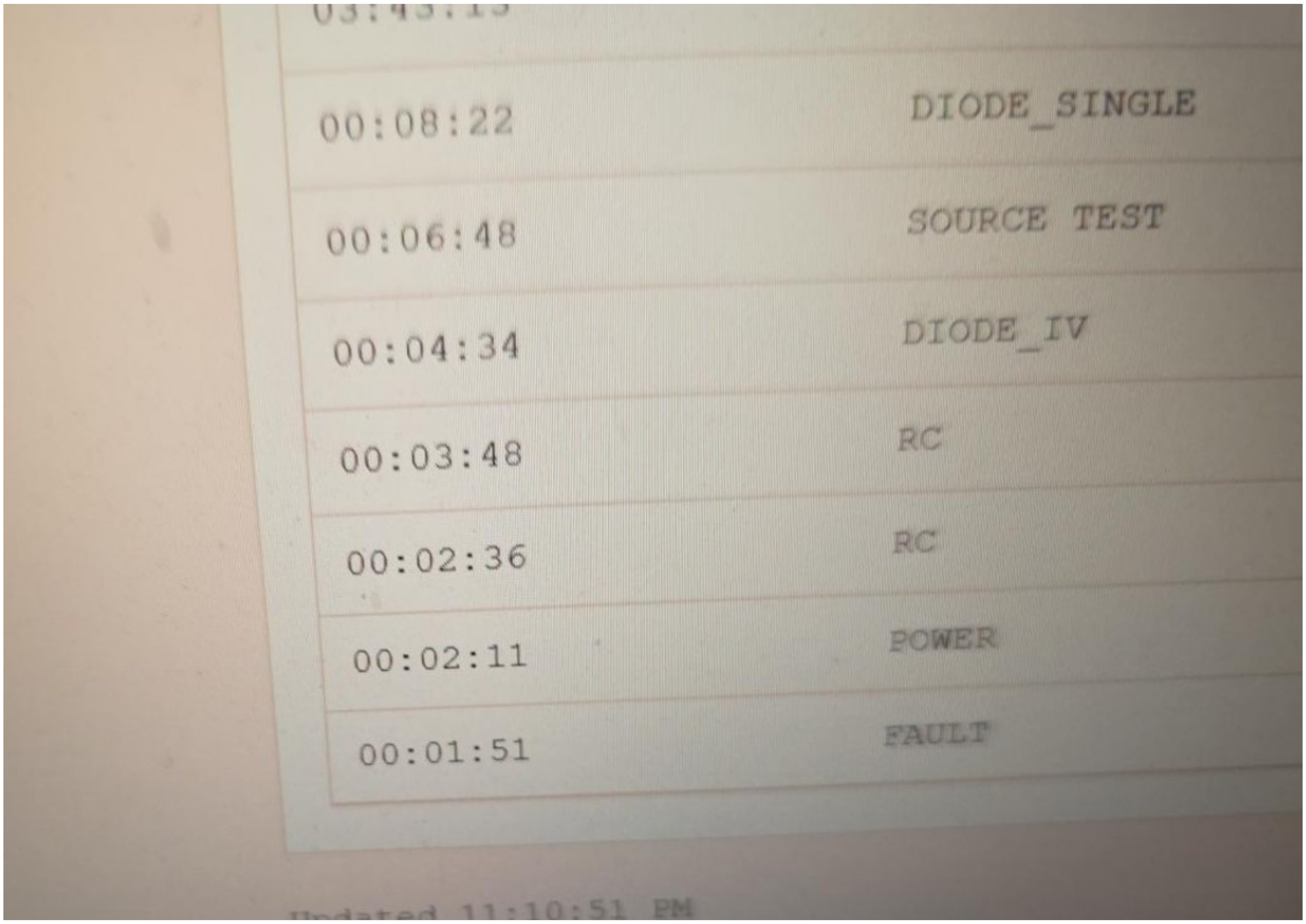
Figure 25



Sanitized dashboard view showing MCU temperature history and the experiment-history table with CSV/JSON export controls.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893841248.jpg.

Figure 26



Sanitized close-up of the experiment-history table and update timestamp.

Privacy treatment: cropped to remove browser/laptop identifiers. Source file: image-1787893852127.jpg.